FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Prematurity & Low Birth Weight

Tutorial | Week 4 · Small babies, sugar & the heart | Neonatology | Dr Nabbale

Preterm birth (before 37 completed weeks) is the leading cause of death in children under 5. Small babies die of hypothermia, hypoglycaemia, infection, feeding failure, apnoea and respiratory distress, and most of these deaths are preventable with warmth (Kangaroo Mother Care), breast milk, infection prevention, caffeine and CPAP.

Must know before the ward round

1. Preterm = born before 37 completed weeks; LBW = birth weight below 2,500 g; VLBW below 1,500 g; ELBW below 1,000 g. A baby can be LBW because she is preterm, small-for-gestational-age (SGA), or both.
2. WHO 2022 - Kangaroo Mother Care (KMC) for ALL preterm/LBW babies, started as soon as possible after birth (immediate KMC), for 8-24 hours a day (as many hours as possible).
3. Normal temperature 36.5-37.5 C; cold stress 36.0-36.4; moderate hypothermia 32.0-35.9; severe below 32. Keep the room at least 25 C.
4. Admit to the special care unit every baby below 2.0 kg or below 35 weeks (WHO); refer very small babies (below 1.5 kg or below 32 weeks) from HC III/IV (UCG 2023).
5. Total fluid/feeds (WHO Pocket Book) - day 1 60 mL/kg/day, day 2 90, day 3 120, then 150 mL/kg/day (up to 180 when tolerating feeds); divide into 2-3-hourly feeds.
6. VLBW (below 1.5 kg) - start enteral feeds on day 1 (10 mL/kg/day or 2-4 mL every 2 h of expressed breast milk) with the rest IV; advance by up to 30 mL/kg/day (WHO 2022).
7. Mother's own milk first, then donor milk, then formula; give feeds by breast, cup, paladai/spoon or nasogastric tube depending on maturity of suck and swallow (usually coordinated from about 32-34 weeks).
8. Caffeine citrate for apnoea of prematurity - 20 mg/kg loading dose (oral or IV over 30 min), then 5 mg/kg once daily from 24 h later (WHO Pocket Book; many units use 5-10 mg/kg/day).
9. RDS - grunting, recession, tachypnoea within 4 h of birth in a preterm baby; treat with CPAP (bubble CPAP 5-8 cm H2O), oxygen to SpO2 90-95%, IV fluids, antibiotics until sepsis excluded.
10. Target SpO2 90-95% in preterm babies on oxygen: too little kills, too much blinds (retinopathy of prematurity) and damages the lungs.
11. Supplements once on full feeds - vitamin D 400 IU daily; iron 2-4 mg/kg/day from 2 weeks to 6 months (WHO); vitamin K 0.5 mg IM at birth (WHO 0.4 mg/kg, max 1 mg) for preterm.
12. Discharge when feeding well (breast/cup), gaining weight on 3 consecutive days, temperature normal for 3 days in KMC/cot, no danger signs, and mother confident; review weekly until 3 kg (WHO).

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1. Definition

A **preterm** baby is born before **37 completed weeks** of gestation. A **low birth weight (LBW)** baby weighs **less than 2,500 g** at birth, whatever the gestation. In Uganda most LBW babies are preterm, but many are **small-for-gestational-age (SGA)** term babies who grew poorly in utero.

Term	Definition
Preterm	Born **before 37 completed weeks** (before 259 days)
Extremely preterm	**<28 weeks**
Very preterm	**28 to <32 weeks**
Moderate to late preterm	**32 to <37 weeks** (late preterm = 34 to 36+6)
Low birth weight (LBW)	Birth weight **<2,500 g**
Very low birth weight (VLBW)	**<1,500 g**
Extremely low birth weight (ELBW)	**<1,000 g**
Small for gestational age (SGA)	Birth weight **below the 10th centile** for gestation and sex (INTERGROWTH-21st or Fenton charts)
Appropriate for GA (AGA)	10th–90th centile
Large for GA (LGA)	Above the 90th centile (think maternal diabetes)
Corrected (adjusted) age	Chronological age minus weeks born early; use it for growth and development until **2 years**
Postmenstrual age (PMA)	Gestational age at birth + chronological age (used for ROP screening and caffeine decisions)
Apnoea	Cessation of breathing for **≥20 seconds**, or shorter with **bradycardia (HR <100)** or desaturation or cyanosis

EXAM FAVOURITE

"Is every LBW baby preterm?" No. A 2.1 kg baby at 39 weeks is LBW but **term and SGA**. Always state both **gestation** and **weight-for-gestation** (for example "32 weeks, 1.45 kg, AGA").

CLINICAL PEARL

The one thing never to forget: a small baby's four killers are **cold, hunger (hypoglycaemia), infection and breathing problems**. On every ward round, ask about each one: temperature, feeds and glucose, signs of sepsis, and apnoea or respiratory distress.

2. Epidemiology

- **Global**: WHO estimates **13.4 million babies were born preterm in 2020** (about **1 in 10**), and **prematurity is the leading cause of death in children under 5** worldwide. Sub-Saharan Africa and South Asia carry most of the burden and most of the deaths.
- **Uganda**: prematurity, intrapartum-related events (asphyxia) and infection are the three leading causes of neonatal death, and small babies are over-represented in all three. Small babies who survive the first week remain at risk of infection, poor growth and neurodevelopmental problems.

- **The good news**: the interventions that save small babies are **simple and cheap**: **warmth (Kangaroo Mother Care)**, **breast milk**, **clean hands and cord care**, **early recognition and treatment of infection**, **safe oxygen and bubble CPAP**, and **caffeine**. The WHO 2022 recommendations put **immediate KMC** and **family involvement** at the centre of care.
- The biggest gains come from things you can do anywhere: keeping the baby warm, starting breast milk on day 1, hand-washing, and referring very small babies **before** they are cold and hypoglycaemic.

Services by level of care Verify *compiled from the source notes, not an MoH equipment list*

Level	Usually available	Often missing
HC III	KMC, cup feeding, EBM, first-dose antibiotics, chlorhexidine, vitamin K	Oxygen, pulse oximeter, IV infusions for neonates
HC IV	Many now have a **KMC unit** and newborn unit with oxygen	CPAP, caffeine, phototherapy at some
General Hospital / RRH	Newborn unit, oxygen, **bubble CPAP** at many RRHs, phototherapy, blood culture at some	Surfactant, ventilation, ROP screening
NRH and PNFP hospitals (Mulago/Kawempe, Mengo, Nsambya, Lubaga)	Bubble CPAP, caffeine, cranial ultrasound; **surfactant**, mechanical ventilation and **ROP laser** in a few centres (mainly Kampala)	Human milk banks (the first opened at Nsambya in 2021)

UGANDA CONTEXT

UCG 2023 asks HC III/IV to look after babies **1.5–2.5 kg** or "up to 1 month early" (extra care, KMC, cup feeding) and to **refer urgently** babies **<1.5 kg or <32 weeks** to hospital, ensuring extra warmth (KMC) during transfer. In practice, Kampala hospitals receive many very small babies from district facilities; a well-warmed baby with a glucose check and a first dose of antibiotics travels much better.

3. Aetiology and Risk Factors

Causes of preterm birth you will see at Mengo, Mulago and Kawempe

Spontaneous	Provider-initiated (iatrogenic)	Background risk factors
Spontaneous preterm labour, often with infection: **UTI, malaria, chorioamnionitis**	**Pre-eclampsia** (delivery for the mother's sake)	**Teenage pregnancy**
PPROM	APH (abruption, praevia)	Poor ANC
Multiple pregnancy	Fetal compromise	Maternal HIV
Cervical insufficiency		Severe anaemia, malnutrition

Causes of SGA (growth restriction): maternal malnutrition, **malaria**, **hypertension**, **HIV**, smoking, **twins**, placental insufficiency, congenital infection or anomaly.

4. Pathophysiology

Why small babies get into trouble

Problem	Why it happens	What you do about it
Hypothermia	Large surface area for weight, thin skin (high evaporative loss), little brown fat and subcutaneous fat, cannot shiver, extended posture	Dry, skin-to-skin/KMC, hat, warm room (≥25 °C), delay bathing
Hypoglycaemia	Small glycogen and fat stores (laid down in the third trimester), high brain-to-body ratio, delayed feeds, cold and sepsis use glucose	Early feeds (within 1 h), 2–3-hourly feeds, glucose checks, IV 10% dextrose if needed
Respiratory distress syndrome (RDS)	**Surfactant deficiency** (type II pneumocytes mature from about 24 weeks, adequate from about 34–35 weeks); alveoli collapse at end-expiration, so the work of breathing rises and oxygenation falls	Antenatal steroids, CPAP, oxygen, surfactant where available
Apnoea of prematurity	Immature brainstem respiratory control; worse with cold, sepsis, hypoglycaemia, anaemia, reflux, feeding	Caffeine, gentle stimulation, monitor, treat the cause
Feeding problems	Suck–swallow–breathe coordination matures at about **32–34 weeks**; small stomach; slow gut motility	NGT or cup feeds of expressed breast milk, slow advancement
Necrotising enterocolitis (NEC)	Immature gut mucosa + formula + ischaemia + bacterial overgrowth	Breast milk, cautious advancement, stop feeds if distension
Infection	Low transplacental IgG (mostly crosses after 32 weeks), thin skin, invasive procedures, long admission	Hand hygiene, KMC, breast milk, minimal lines
Jaundice	Immature liver conjugation, shorter red cell life, slower gut transit	Lower phototherapy thresholds
Intraventricular haemorrhage (IVH)	Fragile germinal matrix vessels (<32 weeks), fluctuating blood pressure	Gentle handling, avoid rapid boluses and hypercapnia
Retinopathy of prematurity (ROP)	Retinal vessels are incomplete; high or fluctuating oxygen drives abnormal vessel growth	Target SpO_2 **90–95%**, screen at-risk babies
Anaemia of prematurity	Low iron stores, blood sampling, low erythropoietin	Iron from 2 weeks, minimise sampling, transfuse if symptomatic
Metabolic bone disease	Calcium and phosphate are mostly transferred in the third trimester	Breast milk, vitamin D, mineral supplements in VLBW

Born early: organs not ready for extrauterine life

↓

Thin skin, little fat, no shivering | Small glycogen stores | Surfactant deficiency

Immature brainstem | Immature suck-swallow, gut | Low maternal IgG

Fragile germinal matrix, incomplete retinal vessels

↓

Hypothermia | RDS: grunting, recession, hypoxaemia | Feeding failure, NEC

Sepsis, meningitis | IVH, PVL, ROP

↓

Hypoglycaemia

↓

Apnoea, bradycardia

↓

Death or disability: CP, BPD, blindness, deafness

Feature links

- Thin, shiny, red skin with lanugo, soft ear cartilage, few sole creases, little breast tissue → **immaturity** (Ballard score).
- Wasted, dry peeling skin, loose skin folds, "old man" face, good sole creases → **SGA (growth restriction)**.
- Surface area, evaporation, no shivering → **cold peripheries, low axillary temperature**.
- Low glycogen and high brain demand → **jitteriness, lethargy, apnoea, seizures** (hypoglycaemia).
- Surfactant deficiency → alveolar collapse → **grunting** (expiration against a closed glottis), **recession, tachypnoea, cyanosis** within 4 h of birth, worsening over 24–48 h.
- Immature respiratory control → **apnoea with bradycardia and desaturation**, starting day 1–3.
- Immature suck → **poor or uncoordinated feeding**; gut ischaemia and formula → **distension, bilious aspirates, blood in stool** (NEC).
- Low IgG and invasive care → **temperature instability, lethargy, new apnoea, feed intolerance** (sepsis).
- Germinal matrix bleed → **bulging fontanelle, falling Hb, seizures** (IVH).
- High or fluctuating oxygen → **ROP** → blindness.
- Low iron and erythropoietin → **pallor, tachycardia, poor weight gain, apnoea** (anaemia).

> **NOTE**
> Antenatal **dexamethasone** (6 mg IM 12-hourly × 4 doses to the mother) induces surfactant production and reduces RDS, IVH, NEC and death when given under the WHO 2022 conditions (24–34 weeks, accurate dating, birth likely within 7 days, no maternal infection, adequate newborn care). See the Preterm Labour, PROM & PPROM topic.

5. Classification / Staging

By gestation: extremely preterm (<28 weeks), very preterm (28 to <32), moderate to late preterm (32 to <37; late preterm 34 to 36+6).

By birth weight: LBW (<2,500 g), VLBW (<1,500 g), ELBW (<1,000 g).

By weight-for-gestation: SGA (<10th centile), AGA (10th–90th), LGA (>90th).

Two kinds of small baby (they behave differently)

Feature	Preterm AGA	Term SGA (growth-restricted)
Appearance	Thin, shiny, red skin, lanugo, soft ear cartilage, few sole creases, little breast tissue	Wasted, dry peeling skin, loose skin folds, "old man" face, alert, good sole creases
Main problems	RDS, apnoea, poor suck, hypothermia, NEC, IVH, jaundice, ROP	**Hypoglycaemia** (low glycogen), **polycythaemia**, hypothermia, asphyxia, meconium aspiration
Feeding	Often cannot suck; NGT or cup	Usually sucks well; hungry

Temperature (axillary, WHO)

Temperature	Classification	Action
36.5–37.5 °C	Normal	Continue KMC or warm care
36.0–36.4 °C	Cold stress (mild)	Skin-to-skin/KMC, warm room, cover head, recheck in 1 h
32.0–35.9 °C	Moderate hypothermia	KMC or incubator/radiant warmer, check glucose, feed, look for sepsis
<32 °C	Severe hypothermia	Rapid rewarming (incubator or radiant warmer), IV 10% dextrose, antibiotics, oxygen, monitor closely

Admission thresholds: admit to the special care unit every baby **<2.0 kg or <35 weeks** (WHO); refer from HC III/IV every baby **<1.5 kg or <32 weeks** (UCG 2023).

6. Clinical Features

History

Question	Why it matters
Gestation: LNMP, early scan, fundal height or Ballard score?	Decides expected problems (<32 weeks: RDS, apnoea, IVH, ROP; 34–36: feeding and jaundice) and the caffeine and ROP plan
Birth weight and current weight	Classifies LBW/VLBW; percentage weight loss guides feeds
Antenatal steroids? How many doses, when? $MgSO_4$?	A full course >24 h before birth lowers RDS risk; $MgSO_4$ <32 weeks is neuroprotective
Why was the baby born early? PPROM, fever, foul liquor, pre-eclampsia, APH, twins	PPROM >18 h, maternal fever >38 °C or foul liquor = **sepsis risk** (antibiotics for at least 48 h, then reassess); pre-eclampsia → SGA, hypoglycaemia, neutropenia
Resuscitation at birth? Apgar	Asphyxia adds HIE, hypoglycaemia, feeding problems
Maternal HIV status, last viral load, syphilis, HBsAg, blood group	Decides NVP or AZT + NVP prophylaxis, syphilis treatment, HepB birth dose; jaundice risk
Feeding: breastfeeding, cup, NGT? Volume and frequency? Any other fluids?	Under-feeding causes hypoglycaemia; prelacteal feeds and formula raise NEC and infection risk
Temperature: how kept warm? KMC hours?	Hypothermia is the commonest, most preventable problem

Question	Why it matters
Apnoeas, colour change, "stops breathing"?	Apnoea of prematurity versus sepsis, hypoglycaemia, seizure
Urine and stool passed?	Hydration, gut function (NEC, obstruction)
Twin?	Other twin's condition; discharge both together (UCG)
Immunisation: BCG, OPV0 (HepB birth dose where available)	WHO: small babies get scheduled vaccines at birth or by discharge
Mother's health and support; distance from facility	KMC needs a well mother or a family member; who will help at home?

Examination

Follow Hutchison's Clinical Methods (24th edition) order adapted for the small newborn: **minimal handling**, keep the baby warm throughout (examine in the incubator, under a radiant warmer or in KMC position where possible), and observe before you touch.

Introduction, consent and preparation

- Introduce yourself to the mother; confirm the baby's name, age in days, gestation and sex; explain and gain consent.
- **Hand hygiene** before and after every touch.
- **Position and exposure**: warm room (≥25 °C); expose only what you need, briefly; return the baby to KMC afterwards.

General inspection (before touching)

Sign	Mechanism	What it means
Activity, cry, posture (extended in very preterm)	Maturity, illness	Lethargy = sepsis, hypoglycaemia, IVH
Colour: pink, pale, blue, yellow, mottled	Anaemia, hypoxaemia, jaundice, poor perfusion	Act on each
Grunting, nasal flaring, intercostal or subcostal recession, tachypnoea	Surfactant deficiency, pneumonia	RDS (Silverman-Andersen score if your unit uses it)
Apnoea or periodic breathing	Immature control, sepsis	Apnoea chart
Skin maturity: thin, shiny, lanugo versus dry, peeling, loose folds	Preterm versus SGA	Classify
Dysmorphism	Syndromes, congenital infection	Cause of SGA

Vital signs and growth: temperature (axillary; **low-reading thermometer**), RR (normal 30–60/min; count a full minute), HR (100–160/min), SpO_2, capillary refill. **Weigh** (naked, same scale), **length**, **head circumference**; plot on a preterm chart (INTERGROWTH-21st or Fenton); classify AGA or SGA; percentage weight loss from birth.

Hands, face and neck

- **Hands**: perfusion, capillary refill, palmar creases.
- **Face**: jaundice (blanch over the nose or sternum), pallor, central cyanosis; ear cartilage (maturity).
- **Head**: fontanelle (bulging = IVH or meningitis; sunken = dehydration), sutures, head circumference.

The systems

- **Respiratory**: inspection (recession, flaring), auscultation (air entry; asymmetry suggests pneumothorax).
- **Cardiovascular**: murmur (PDA), femoral pulses, perfusion.

- **Abdomen**: distension, tenderness, visible loops, discoloration, bowel sounds (NEC); cord (chlorhexidine, signs of infection).
- **Skin**: jaundice, pustules, pressure areas.
- **Neurology**: tone, posture, activity, suck, fontanelle (IVH, meningitis).
- **Maturity**: **Ballard** features (skin, lanugo, plantar creases, breast bud, ear, genitalia; posture, square window, scarf sign). See the Examination of the Newborn topic.

Completing the examination

- **Thank** the mother, return the baby to KMC or the incubator, **hand hygiene**, and **summarise**: gestation, weight-for-gestation, temperature, respiratory support, feeds, glucose, sepsis risk.
- **Document negatives**: "no grunting, no apnoeas in 24 h, abdomen soft, no jaundice".

Red flags

RED FLAG

- Temperature **<35.5 °C** or >37.5 °C, or a baby who stays cold despite KMC
- **Apnoea** (pause >20 s, or shorter with bradycardia or cyanosis), especially recurrent or new after day 3 (think sepsis)
- **Grunting, severe chest indrawing, SpO_2 <90%**, or rising oxygen need
- **Not feeding**, vomiting, **abdominal distension, bilious aspirates, blood in stool** (NEC)
- **Lethargy, floppy, convulsions**, bulging fontanelle
- **Jaundice on day 1** or yellow palms and soles
- **Glucose <2.6 mmol/L**, especially if recurrent
- Weight loss **>10–15%** or no weight gain after day 7–10

Thinking like a clinician

- **A sick preterm baby has sepsis until proven otherwise.** Apnoea, temperature instability, feed intolerance and lethargy are non-specific. Check glucose and temperature, then start **ampicillin + gentamicin** after a blood culture where possible.
- **New apnoea after the first few days, in a baby who was stable, is not "just prematurity".** Think sepsis, NEC, meningitis, hypoglycaemia, anaemia, IVH.
- **Respiratory distress in a preterm baby**: within 4 h of birth and worsening over 24–48 h = **RDS**; with maternal fever or PPROM = **pneumonia or sepsis** (cannot be separated clinically, so treat both); in a late preterm born by CS without labour = **transient tachypnoea**; sudden collapse on CPAP = **pneumothorax**.
- **An SGA term baby is a different animal**: feed early and check glucose; he is at risk of hypoglycaemia and polycythaemia, not RDS.
- **Do not chase weight before warmth.** A cold baby does not gain weight and does not tolerate feeds. Fix temperature first.
- **Count fluids as total fluids** (IV + feeds). Students often add full IV fluids on top of feeds and drown the baby.
- **Oxygen is a drug.** Use a pulse oximeter, target **90–95%**, wean as soon as possible.

7. Differential Diagnoses

The small baby with respiratory distress

Differential	Distinguishing clinical feature	Distinguishing investigation
RDS	Preterm, onset within 4 h, worsening over 24–48 h, grunting	CXR: ground-glass, air bronchograms
Transient tachypnoea of the newborn	Late preterm or term, CS without labour, mild, improves in 24–72 h	CXR: fluid in fissures, hyperinflation
Pneumonia / sepsis	PPROM, maternal fever, foul liquor; temperature instability	Culture, CRP, FBC; CXR patchy
Pneumothorax	Sudden deterioration (often on CPAP), asymmetric chest	Transillumination, CXR
Congenital heart disease	Cyanosis out of proportion to distress, murmur, poor pulses	Pre- and post-ductal SpO_2, echocardiography
Hypothermia / hypoglycaemia	Cold, jittery, tachypnoea that settles with warmth or glucose	Temperature, glucose

The small baby with apnoea: apnoea of prematurity (a diagnosis of exclusion, <34 weeks, starts day 1–3), sepsis or meningitis, hypoglycaemia, hypothermia or hyperthermia, seizures, IVH, NEC, anaemia, reflux or feeding, maternal drugs ($MgSO_4$, opioids).

Preterm versus SGA: plot weight against gestation (INTERGROWTH-21st or Fenton); <10th centile = SGA; check maturity with Ballard.

8. Clinical Workup

"Available at" is the lowest level that usually offers the test Verify *availability compiled from the source notes*.

Bedside

Test	Why	Expected / abnormal finding	Available at
Glucose (glucometer)	Commonest treatable cause of lethargy, apnoea, seizures	<2.6 mmol/L = treat; check at 1–2 h of age, then before feeds 3–6-hourly until stable on feeds	HC III and above
Temperature (low-reading thermometer)	Drives apnoea, hypoglycaemia, acidosis	Hypothermia grade	All levels
Pulse oximetry	Oxygen decisions	Target 90–95% on oxygen; <90% in air = oxygen	HC IV and above
Weight daily (same scale)	Fluid and feed decisions	Losses up to 10–15% in VLBW in week 1; regain birth weight by 10–14 days (later in VLBW); then about **15 g/kg/day**	All levels
Gastric aspirate before feeds (if NGT)	Feed tolerance	Bilious or more than half the previous feed = feed intolerance (think NEC or sepsis)	All levels
Transcutaneous bilirubin (if available)	Jaundice screening	Above gestation-specific threshold → serum	Some hospitals
HemoCue Hb / PCV	Anaemia or polycythaemia	Anaemia (<10–12 g/dL symptomatic); polycythaemia in SGA (venous PCV >65%)	HC IV and above

Test	Why	Expected / abnormal finding	Available at
Ballard score	Gestation when dates are unknown	Gestation in weeks	All levels

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
FBC	Suspected sepsis, pallor, SGA	WBC <5 or >20 × 10^9/L, platelets <100, Hb, PCV	Supports sepsis; transfusion; partial exchange for polycythaemia	HC IV and above
Serum bilirubin	Visible jaundice, especially day 1 or preterm	Compare with gestation-specific thresholds	Phototherapy or exchange	Referral hospitals
CRP	Sepsis work-up	>10 mg/L at 18–24 h supports infection	Two normal CRPs help stop antibiotics at 36–48 h	Referral hospitals, private labs
Blood culture	Before antibiotics in any suspected sepsis	Organism and sensitivities (*Klebsiella*, *S. aureus*, *E. coli* common)	Narrow or change antibiotics; duration	Regional and national referral, Mengo, Nsambya
Electrolytes, calcium	VLBW on IV fluids, seizures	Hypo- or hypernatraemia, hypocalcaemia	Adjust fluids; calcium gluconate	Referral hospitals

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Chest X-ray	Respiratory distress	RDS: ground-glass, air bronchograms, low volumes; pneumothorax; pneumonia	CPAP or surfactant; chest drain	Most hospitals
Abdominal X-ray	Suspected NEC	Pneumatosis intestinalis, portal gas, free air	Nil by mouth, antibiotics, surgical referral	Most hospitals
Cranial ultrasound	<32 weeks, seizures, falling Hb	IVH, PVL	Prognosis, follow-up	Kampala tertiary centres

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Lumbar puncture	Sepsis with seizures, bulging fontanelle, or positive culture	Raised WBC, low glucose	Meningitis = 21 days IV antibiotics	Hospital
ROP eye examination	At-risk babies (section 9)	Stage, zone, plus disease	Laser or anti-VEGF	Few centres (Kampala)
Hearing screen	After jaundice, gentamicin, meningitis	Hearing loss	Early intervention	Referral centres **Verify** *availability*

Putting it together

Model one-line summary

> "Baby girl of Nakato, day 3 of life, born at **32 weeks** (by early scan) to a 19-year-old para 1 after preterm labour with **ROM for 20 hours**; mother received 2 doses of dexamethasone. Birth weight **1.45 kg** (VLBW, AGA), today **1.38 kg** (5% loss). She is in KMC, temperature 36.6 °C, RR 58 with mild recession on bubble CPAP 5 cm H_2O in 25% oxygen, SpO_2 93%, had two apnoeas yesterday, and is on EBM 4 mL 2-hourly by NGT plus IV 10% dextrose. Mother HIV-negative at ANC, syphilis negative; BCG and OPV0 given."

Problem list

1. VLBW very preterm infant.
2. RDS on CPAP.
3. Apnoea of prematurity versus sepsis.
4. Risk of early-onset sepsis (ROM 20 h) on ampicillin + gentamicin, culture pending.
5. Establishing enteral feeds.
6. Thermal care in KMC.
7. Jaundice and ROP screening plans.
8. Family support.

Model case presentation (ward round)

> "This is **Baby Nakato**, a **9-day-old** female born at **31+2 weeks** by early scan, now **32+4 weeks PMA**. Birth weight **1.30 kg (VLBW, AGA)**, today **1.27 kg**, having regained 40 g in the last 3 days. She was born after preterm labour with **ROM for 22 hours**; mother had **one dose of dexamethasone**. She needed **bag-mask ventilation for 1 minute**, Apgar 5 and 8.
>
> She had **RDS**, treated with **bubble CPAP for 4 days**, now in **room air** with SpO_2 95%. She is on **caffeine citrate 5 mg/kg daily**, last apnoea 3 days ago. Blood culture grew nothing; she completed **5 days of ampicillin and gentamicin**. She is on **EBM 150 mL/kg/day by NGT**, 24 mL 3-hourly, with some cup feeds, no aspirates. She is in **KMC 20 h/day**, temperature 36.6–37.2 °C. Phototherapy for 2 days for jaundice, now stopped.
>
> Mother is **HIV-negative** (retested in labour), RPR negative. **BCG and OPV0 given.** Weight-for-age on the INTERGROWTH chart is on the 25th centile.
>
> **Assessment:** a stable, growing VLBW very preterm infant; problems: (1) feeding progression; (2) apnoea of prematurity on caffeine; (3) needs ROP screening at 4 weeks; (4) anaemia and bone health supplementation; (5) discharge planning with KMC at home.
>
> **Plan:** advance to cup feeds, start iron at 2 weeks and vitamin D, continue caffeine to 34–35 weeks PMA, ROP referral, check Hb weekly, discharge when criteria met."

Worked example 1: feeds. A 1.6 kg baby on day 3 needs **120 mL/kg/day × 1.6 = 192 mL/day**. Given 2-hourly (12 feeds) that is **16 mL per feed**; 3-hourly (8 feeds) **24 mL per feed**. If she is also on IV fluids, subtract the feed volume from the total: for example feeds 10 mL × 12 = 120 mL, so IV = 192 − 120 = **72 mL/day = 3 mL/h**.

Worked example 2: feeds on day 4. A 1.6 kg baby on day 4: 150 mL/kg/day = 240 mL/day = **30 mL 3-hourly**.

Worked example 3: WHO feed volumes. A 1 kg baby on 150–180 mL/kg/day = **19–23 mL 3-hourly**; a 1.5 kg baby = **28–34 mL 3-hourly**.

9. Management

Immediate / emergency

At birth and in the first 5 minutes

1. **At birth**: dry, keep warm, assess breathing; resuscitate per HBB (size **0** mask for small babies; room air or 21–30% oxygen below 35 weeks). Delayed cord clamping (at least 1 min) if breathing.
2. **Temperature** now. If <36.5 °C, start **skin-to-skin/KMC** (or incubator or radiant warmer) and recheck in 30–60 min.
3. **Breathing**: count RR, look for grunting and recession, check **SpO_2**. If distressed: oxygen by nasal prongs to SpO_2 **90–95%**, consider **bubble CPAP**.
4. **Glucose**: check at **1–2 h of age** (or now if unwell). If <**2.6 mmol/L**: feed or give **10% dextrose 2 mL/kg IV**.
5. **Danger signs** (not feeding, convulsions, lethargy, temperature <35.5 °C or >37.5 °C, RR >60, severe indrawing, cyanosis, apnoea): admit, IV access, **antibiotics**, call for senior help.
6. **<1.5 kg or <32 weeks at HC III/IV**: stabilise (warmth, glucose or feed, first antibiotic dose if indicated) and **refer in KMC** with a note.

Emergency flow for the small baby on admission (ABCDE)

1. **Warm (E first in a small baby)**: KMC or incubator or radiant warmer; hat, socks; room ≥25 °C.
2. **Airway and breathing**: position, gentle suction only if needed; oxygen by nasal prongs (0.5–1 L/min) to SpO_2 90–95%; **bubble CPAP** for RDS (start 5–6 cm H_2O, up to 7–8).
3. **Circulation**: CRT, HR; IV access (peripheral or umbilical vein).
4. **Glucose**: if <2.6 mmol/L give **10% dextrose 2 mL/kg IV** over 5 min, then maintenance; recheck in 30 min.
5. **Antibiotics** if any sepsis risk or danger sign: **ampicillin + gentamicin**.
6. **Vitamin K** if not given; **chlorhexidine** cord care; **tetracycline eye ointment** if not given.
7. **Feeds**: start EBM as early as possible (day 1), even small volumes.
8. **Caffeine** for babies <34 weeks with apnoea (and to consider for prevention).
9. **Document and communicate**: weight, plan, family counselling.

Definitive

Thermal care and Kangaroo Mother Care

How to do KMC (WHO 2022)

1. Baby naked except nappy, hat and socks, placed **upright, skin-to-skin between the mother's breasts**, head turned to one side, slightly extended.
2. Secure with a binder or wrap; cover mother and baby with her clothes or a blanket.
3. Continue **day and night** (at least **8–24 h/day**, as many hours as possible); fathers and relatives can take turns.
4. Start **immediately after birth** (iKMC) once the baby can tolerate handling; WHO 2022 now recommends this even for babies still needing oxygen or CPAP where the unit can support it.
5. Breastfeed in the KMC position; monitor temperature 4–6-hourly at first.
6. Mother sleeps propped up at 15–30 degrees.

KMC = continuous skin-to-skin contact plus exclusive breastfeeding plus early discharge with follow-up.

> **CLINICAL PEARL**
> KMC reduces mortality, hypothermia, sepsis and length of stay and improves breastfeeding and bonding. It is not a "second-best incubator"; for stable small babies it is **better** than an incubator.

Feeding the small baby

Method by maturity (approximate)

Gestation / weight	Usual feeding method
≥34 weeks / usually ≥2.0 kg	**Breastfeed** directly; support positioning
32–34 weeks / about 1.5–2.0 kg	Breast + **cup or paladai** with expressed breast milk (EBM)
<32 weeks / below about 1.5 kg	**Nasogastric or orogastric tube** EBM ± IV fluids; move to cup or breast as suck matures
Very sick, distended abdomen, shocked, frequent seizures	**Nil by mouth**, IV fluids; trophic feeds once stable

Fluid and feed volumes (total = IV + enteral) (WHO Pocket Book 2013)

Day of life	Total fluids (mL/kg/day)
Day 1	**60**
Day 2	**90**
Day 3	**120**
Day 4 onwards	**150** (up to **180** after some days if tolerating feeds)

- Give feeds **every 2–3 hours** (8–12 feeds a day).
- **<1.5 kg**: on day 1 start **10 mL/kg/day** EBM (or 2–4 mL every 2 h by NGT if well) with the remaining ~50 mL/kg/day as IV 10% dextrose. Increase feeds by **1–2 mL per feed per day** (Pocket Book) or **up to 30 mL/kg/day** (WHO 2022). Aim for full enteral feeds within **5–7 days** so the drip can come out.
- **IV fluid**: 10% dextrose for the first 2 days, then a sodium-containing solution (for example 10% dextrose with 0.18% saline, or half-strength Darrow's per unit protocol). **Do not exceed 100 mL/kg/day IV** unless dehydrated or under phototherapy or a radiant heater (Pocket Book). Use a burette; check hourly.
- Add **10–20%** to fluids under phototherapy or radiant warmers.
- **Feed intolerance**: vomiting, distension, aspirates more than half of the feed or bilious → reduce or stop feeds, examine for NEC and sepsis.
- **Milk**: **mother's own milk** first; **donor human milk** if not available (where a milk bank exists); **preterm formula** only if neither (WHO 2022). Help the mother express by hand within 1 h of birth and every 2–3 h (8 times a day) to build supply.
- Weight gain target after regaining birth weight: about **15–20 g/kg/day**.

> **UGANDA CONTEXT**
> Uganda has few human milk banks (the first opened at Nsambya in 2021), so in most units **mother's own EBM** is the norm, with formula used reluctantly when the mother is very ill or absent. Expressing support (privacy, a chair, a clean cup, praise) is a real clinical intervention.

Apnoea of prematurity and caffeine

- Common below **34 weeks**, especially below 30 weeks; typically starts day 1–3 and resolves by **34–37 weeks PMA**.
- Immediate response: **gentle stimulation** (rub the back or soles); if no breathing, **bag-mask ventilation**; check glucose and temperature; consider sepsis.
- Monitor with a pulse oximeter with the alarm on, or an apnoea monitor.
- **Caffeine citrate** (first choice) until **34–35 weeks PMA** and **5–7 days apnoea-free** (WHO: 4–5 days after apnoea stops); **aminophylline** if caffeine is unavailable (drug table).
- WHO 2022: caffeine is **recommended for treatment** of apnoea in preterm infants, **for extubation** below 34 weeks, and **may be considered for prevention** of apnoea below 34 weeks.

Respiratory distress syndrome and CPAP

1. **Minimal handling**, thermal care, nil by mouth or trophic feeds initially; IV fluids.
2. **Oxygen** by nasal prongs to keep SpO_2 **90–95%** (not higher).
3. **Bubble CPAP** (WHO 2022 strong recommendation for RDS; may be considered immediately after birth for babies <32 weeks): start **5–6 cm H_2O**, FiO_2 to target SpO_2, nasal prongs of the right size, NGT on free drainage for air, protect the nasal septum.
4. **Antibiotics** (ampicillin + gentamicin) because pneumonia cannot be excluded; stop at 48 h if culture negative and baby well.
5. **Surfactant** (via ETT, or less-invasive methods) where available, usually if FiO_2 needs to exceed about 0.30–0.40 on CPAP.
6. **Escalate** if apnoea, rising FiO_2 above 0.4–0.6, worsening work of breathing, or pneumothorax: refer for ventilation where possible.
7. Expect improvement after **48–72 h** as the baby makes surfactant.

> **UGANDA CONTEXT**
> Bubble CPAP made from locally available materials is used in many Ugandan units. The main dangers are **unblended 100% oxygen** (ROP, lung injury) and **nasal trauma**. Use an oxygen blender or air-oxygen mixing and a pulse oximeter.

Infection prevention and treatment

- **Hand hygiene** before and after every touch; mothers and visitors too. KMC and breast milk reduce infection.
- **Chlorhexidine 7.1% digluconate** to the cord (Uganda policy) and nothing else.
- Antibiotics for risk factors or signs (drug table; see also Common Neonatal Disorders): stop at 48 h if well and cultures negative; 7–10 days for sepsis.

Jaundice in the preterm

Preterm babies get jaundice more often, later (peak day 5–7), and it becomes dangerous at lower bilirubin levels. Use **gestation-specific thresholds** (NICE CG98 graphs; WHO Pocket Book: lower thresholds "for preterm below 35 weeks or any risk factor"). Phototherapy for any visible jaundice on day 1. See Common Neonatal Disorders.

Supplements and routine care

Supplement	Dose	Route	When / duration
Vitamin K	**0.5 mg** (Uganda/local); WHO **0.4 mg/kg** (max 1 mg) for preterm; 1 mg term	IM	Once at birth
Vitamin D	**400 IU**	Oral	Daily once on full feeds (WHO 2022: may be considered)
Iron (ferrous sulphate, elemental iron)	**2–4 mg/kg/day elemental iron**	Oral	From **2 weeks** (or when on full feeds) until **6 months** (WHO)
Multivitamins / calcium and phosphate	Per unit protocol (Pocket Book: Ca 120–140 mg/kg, P 60–90 mg/kg/day for VLBW)	Oral	Once on full feeds
Folic acid	Per unit protocol	Oral	Commonly given in Ugandan units
Emollient / topical oil	Thin layer (WHO 2022: may be considered)	Skin	Daily
Probiotics	WHO 2022: may be considered for human-milk-fed babies <32 weeks	Oral	Where a quality product is available

Immunisation: give **BCG and OPV0 at birth or before discharge** regardless of weight (WHO: small babies receive scheduled vaccines on time by chronological age), then the UNEPI schedule from **6 weeks chronological age** (not corrected age).

Retinopathy of prematurity (ROP)

- **Who to screen**: international criteria are **<31–32 weeks or <1.5 kg**. In LMICs (including Uganda) babies up to **34 weeks or 2.0 kg** develop severe ROP when oxygen is unmonitored, so many programmes use **wider criteria**, plus any baby who had **unmonitored oxygen**.
- **When**: first examination at **4 weeks after birth or 31 weeks PMA, whichever is later**, then 1–2-weekly until vessels mature.
- **Prevention**: **SpO_2 90–95%**, blenders, avoid fluctuations, good nutrition, treat sepsis.
- **Treatment**: laser photocoagulation or intravitreal anti-VEGF for treatment-requiring ROP, ideally within 48–72 h of diagnosis.

> **UGANDA CONTEXT**
> ROP screening in Uganda is limited to a few Kampala centres and outreach programmes. If you are in a unit without screening, **write the plan in the discharge summary** ("ROP screen due at 4 weeks") and refer the baby to the nearest eye unit.

Drug table

Drug	Dose (per kg; neonate)	Route	Frequency	Duration	Key cautions
10% dextrose	**2 mL/kg** bolus over 5 min for glucose <2.6 mmol/L; infusion per total fluid table	IV	Bolus then continuous	Until on full feeds	Max about 100 mL/kg/day IV unless dehydrated or under phototherapy; recheck glucose in 30 min
Caffeine citrate (first choice)	Loading **20 mg/kg**; maintenance **5 mg/kg** (may increase by 5 mg/kg steps to max **20 mg/kg/day**; many units use 5–10 mg/kg/day)	Oral or IV over 30 min	Maintenance **once daily**, starting 24 h after the load	Until **34–35 weeks PMA** and **5–7 days apnoea-free**	Tachycardia (>180), jitteriness, feed intolerance. 20 mg caffeine citrate = 10 mg caffeine base
Aminophylline (if caffeine unavailable)	Loading **6 mg/kg** IV over 20 min; maintenance **2.5 mg/kg** (week 1), 4 mg/kg (weeks 2–4)	IV / oral	**12-hourly**	As caffeine	Narrower safety margin: tachycardia, vomiting, seizures
Ampicillin	**50 mg/kg** (100 mg/kg/dose if meningitis)	IV / IM	**12-hourly** in week 1; **8-hourly** weeks 2–4	Stop at 48 h if well and culture-negative; 7–10 days for sepsis; 21 days meningitis	With gentamicin
Gentamicin	**3 mg/kg** (LBW, WHO; UCG 4 mg/kg preterm); **5 mg/kg** (≥2.5 kg) in week 1; **7.5 mg/kg** weeks 2–4	IV / IM	**Once daily** (every 24 h; some units 36–48-hourly below 30 weeks)	As ampicillin	Nephrotoxic, ototoxic; hearing follow-up
Oxygen	Nasal prongs 0.5–1 L/min, or via CPAP	Inhaled	Continuous	Wean as soon as possible	Target SpO_2 **90–95%**; blend with air; hyperoxia → ROP

Drug	Dose (per kg; neonate)	Route	Frequency	Duration	Key cautions
Bubble CPAP	Start **5–6 cm H_2O** (up to 7–8)	Nasal prongs	Continuous	Usually 48–72 h or more for RDS	Nasal trauma, pneumothorax, gastric distension (NGT on free drainage)
Surfactant	Per product	Endotracheal or less-invasive	Usually once, may repeat	n/a	Few centres; if FiO_2 >0.30–0.40 on CPAP
Vitamin K	**0.5 mg** (local); WHO **0.4 mg/kg**, max 1 mg	IM	Once	At birth	
Vitamin D	**400 IU**	Oral	Daily	Once on full feeds	
Iron (elemental)	**2–4 mg/kg/day**	Oral	Daily	**2 weeks to 6 months**	Constipation, dark stools
Dexamethasone (to the mother)	**6 mg**	IM	12-hourly × 4 doses	One course	Only under WHO 2022 conditions (24–34 weeks, accurate dating, birth likely within 7 days, no maternal infection, adequate newborn care)

Referral criteria

Level	What to do	Refer up when
Home / VHT	KMC, exclusive breastfeeding, recognise danger signs	Any danger sign; baby not feeding or cold despite KMC
HC III / HC IV (UCG: babies 1.5–2.5 kg or up to 1 month early)	Extra care, KMC, cup feeding, glucose check, first-dose antibiotics if indicated	**<1.5 kg or <32 weeks**: refer urgently in KMC with a note (UCG 2023); any danger sign; respiratory distress needing oxygen or CPAP
General Hospital / RRH (WHO: admit every baby <2.0 kg or <35 weeks)	SCBU, oxygen, bubble CPAP, IV fluids, caffeine, phototherapy, antibiotics	Rising FiO_2 (>0.4–0.6) on CPAP, recurrent apnoea, pneumothorax, NEC needing surgery, need for surfactant or ventilation → NRH or tertiary
NRH and tertiary (Kampala)	Surfactant, ventilation, cranial ultrasound, ROP screening and laser, paediatric surgery	n/a

Monitoring and follow-up

- **Ward monitoring**: temperature 4–6-hourly in KMC at first; RR, HR, SpO_2 (alarm on); apnoea chart; glucose before feeds 3–6-hourly until stable; daily weight; gastric aspirates; intake and output; jaundice; Hb weekly in VLBW.
- **Discharge criteria (WHO/UCG)**:
  - No danger signs or signs of infection.
  - **Feeding well** on breast (or cup), gaining weight on **3 consecutive days**.
  - Temperature normal in KMC or cot for **3 consecutive days**.
  - **Mother confident** to care for the baby (KMC, feeding, danger signs).
  - Vaccines due given; HIV prophylaxis plan clear.
- **Follow-up**:

- **Weekly** weight and feeding review until **3 kg** (WHO); continue KMC at home until the baby wriggles out (usually around 2.5 kg or 40 weeks PMA).
- **Growth** on preterm charts then WHO charts using **corrected age** to 2 years; weight-for-age, length-for-age, weight-for-length Z-scores.
- **Development** (corrected age): cerebral palsy, hearing (especially after jaundice, gentamicin, meningitis), vision (ROP, squint).
- **Anaemia**: Hb at 6–8 weeks; continue iron.
- **Immunisation** on chronological age.
- Teach **danger signs**: not feeding, cold, fast or difficult breathing, apnoea, convulsions, yellow palms and soles, fever.

10. Complications

Disease-related

Short-term	Long-term
Hypothermia, hypoglycaemia	Cerebral palsy, cognitive and learning difficulties
RDS, pneumothorax, apnoea	**Bronchopulmonary dysplasia** (oxygen need at 36 weeks PMA)
Sepsis, meningitis	**ROP** → blindness
NEC, feeding intolerance	Hearing loss
Jaundice, kernicterus	Growth faltering, metabolic bone disease
IVH, PVL	Anaemia of prematurity
PDA	Behavioural problems, re-admissions
Electrolyte imbalance	Family stress, maternal depression

Treatment-related

Complication	Cause	Recognition and action
ROP, lung injury	Unblended or unmonitored oxygen	SpO_2 90–95%; blender; screen
Fluid overload	Full IV fluids on top of feeds	Count total fluids; IV = total – feeds
Nasal septum injury, pneumothorax	CPAP	Correct prongs, nasal care; sudden deterioration → transilluminate, CXR
NEC	Formula, rapid feed advancement	Breast milk; cautious advancement; stop feeds if distension
Tachycardia, jitteriness, feed intolerance	Caffeine or aminophylline	Reduce dose; aminophylline has a narrower margin (seizures)
Ototoxicity, nephrotoxicity	Gentamicin	Correct dose and interval; hearing follow-up
IVH	Rapid boluses, hypercapnia, rough handling	Gentle handling; avoid rapid boluses
Hospital-acquired infection	Lines, long stay	Hand hygiene; KMC; early enteral feeds so the drip can come out

11. Prognosis

- Survival rises steeply with gestation and birth weight; in Uganda, very preterm and VLBW babies have high mortality, mostly from hypothermia, infection, RDS and feeding failure Verify *local survival figures by gestation*.
- Most deaths are preventable with KMC, breast milk, infection prevention, caffeine and CPAP.
- Long-term risks rise as gestation falls: cerebral palsy, cognitive and learning difficulties, BPD, ROP and blindness, hearing loss, growth faltering.
- Use **corrected age** to judge growth and development until 2 years; many preterm children catch up.

12. Prevention

- **Prevent preterm birth**: good ANC, treat UTI and malaria (IPTp), manage pre-eclampsia, family planning (avoid teenage pregnancy and short intervals), nutrition, HIV care. See Preterm Labour, PROM & PPROM.
- **Improve outcome when preterm birth is likely**: antenatal **dexamethasone** under WHO 2022 conditions; **$MgSO_4$** for neuroprotection <32 weeks; antibiotics for PPROM; deliver or transfer in utero to a facility with newborn care.
- **Prevent complications after birth**: immediate KMC, breast milk from day 1, hand hygiene, chlorhexidine cord care, SpO_2 90–95%, caffeine, CPAP.
- **Prevent late harm**: ROP screening, hearing screening, developmental follow-up, iron and vitamin D.

13. Special Populations

- **Term SGA babies**: feed early and check glucose; risk of hypoglycaemia and polycythaemia (venous PCV >65%), not RDS.
- **Twins**: check the other twin; discharge both together (UCG).
- **HIV-exposed small babies**: NVP or AZT + NVP prophylaxis according to the mother's viral load; give by NGT if unable to suck.
- **Babies of mothers with pre-eclampsia**: SGA, hypoglycaemia, neutropenia.
- **Very preterm (<32 weeks) and VLBW**: IVH, ROP screening, caffeine, metabolic bone disease supplements, slower feed advancement.

14. Key Points

High-yield points

- State gestation **and** weight-for-gestation: preterm, SGA, or both.
- The four killers: cold, hunger (hypoglycaemia), infection, breathing problems.
- Immediate KMC for all stable small babies, 8–24 h a day.
- Temperature 36.5–37.5 °C; room ≥25 °C.
- Total fluids 60/90/120/150 mL/kg/day; IV = total − feeds.
- Mother's own milk first; start EBM on day 1.
- Caffeine citrate 20 mg/kg load, then 5 mg/kg daily.
- Bubble CPAP 5–6 cm H_2O for RDS; SpO_2 90–95%.
- A sick preterm baby has sepsis until proven otherwise.
- Vaccinate by chronological age.
- Discharge: feeding, 3 days of weight gain and normal temperature, confident mother; weekly review until 3 kg.

Numbers & doses

Item	Value
Preterm	**<37 completed weeks**; very preterm 28–32; extremely preterm <28
LBW / VLBW / ELBW	**<2,500 / 1,500 / 1,000 g**
Admit to SCBU (WHO)	**<2.0 kg or <35 weeks**
Refer from HC III/IV (UCG)	**<1.5 kg or <32 weeks**
Normal temperature	**36.5–37.5 °C** (moderate hypothermia 32.0–35.9; severe <32)
Room temperature	**≥25 °C**
KMC	**8–24 h/day**, start **as soon as possible after birth** (WHO 2022)
Total fluids	**60 / 90 / 120 / 150** mL/kg/day on days 1/2/3/4+; up to 180
Max IV fluids	About **100 mL/kg/day** (unless dehydrated, phototherapy or radiant heater)
VLBW feeds day 1	**10 mL/kg/day** EBM (or 2–4 mL 2-hourly); advance up to **30 mL/kg/day**
Glucose threshold	Treat **<2.6 mmol/L**: **10% dextrose 2 mL/kg IV**
Caffeine citrate	**20 mg/kg** load, **5 mg/kg/day** maintenance (max 20)
Aminophylline	**6 mg/kg** load, **2.5 mg/kg 12-hourly** (week 1)
SpO_2 target	**90–95%**
CPAP starting pressure	**5–6 cm H_2O** (up to 7–8)
Iron	**2–4 mg/kg/day** from **2 weeks to 6 months**
Vitamin D	**400 IU/day**
Vitamin K (preterm)	**0.5 mg IM** (WHO 0.4 mg/kg, max 1 mg)
Weight gain target	**15–20 g/kg/day**
ROP first screen	**4 weeks after birth or 31 weeks PMA**, whichever is later
Discharge	Feeding well, weight gain **3 days**, temperature normal **3 days**, mother confident
Follow-up	**Weekly until 3 kg**; corrected age for growth and development to 2 years

Exam traps

- **"Put the cold small baby in a bath of warm water or under a bulb."** → Wrong. Use **KMC** or an incubator or radiant warmer; do not bathe small babies.
- **"A 1.4 kg baby should go to the incubator; KMC is for when he is bigger."** → Wrong. WHO 2022: **immediate KMC** for all stable small babies; it reduces death.
- **"Give IV fluids at 150 mL/kg/day plus full feeds."** → Wrong. Count **total** fluids; IV = total − feeds.
- **"Keep SpO_2 at 100% to be safe."** → Wrong. Target **90–95%**; hyperoxia causes **ROP** and lung injury.
- **"Apnoea on day 6 in a stable 30-weeker is just prematurity; increase caffeine."** → Wrong. **Rule out sepsis** (and NEC, hypoglycaemia, anaemia) first.
- **"Give formula because the mother's milk has not come in."** → Wrong. Colostrum is enough in small volumes; help her **express**; formula only when no mother's or donor milk.
- **"Vaccinate preterm babies at their corrected age."** → Wrong. Use **chronological age**; give BCG and OPV0 before discharge.
- **"SGA term baby: watch for RDS."** → The bigger risk is **hypoglycaemia** (and polycythaemia).

Questions seniors ask

Viva questions

1. *Define LBW, VLBW and ELBW.* <2,500 g, <1,500 g, <1,000 g at birth.
2. *Why do preterm babies get cold?* Large surface-area-to-weight ratio, thin skin with high evaporative loss, little brown and subcutaneous fat, extended posture, cannot shiver.
3. *What is KMC and who should get it?* Continuous skin-to-skin contact plus exclusive breastfeeding plus early discharge with follow-up. WHO 2022: all preterm and LBW babies, started as soon as possible after birth, 8–24 h a day.
4. *How much milk does a 1.6 kg baby need on day 4?* 150 mL/kg/day = 240 mL/day = 30 mL 3-hourly.
5. *What is the caffeine regimen?* Caffeine citrate 20 mg/kg load, then 5 mg/kg daily (up to 20 mg/kg/day) from 24 h later, until about 34–35 weeks PMA and 5–7 days apnoea-free.
6. *What SpO_2 do you target in a preterm baby?* 90–95%.
7. *Name four signs of NEC.* Abdominal distension, feed intolerance or bilious aspirates, blood in stool, abdominal wall discoloration; plus apnoea, lethargy, temperature instability.
8. *When can a small baby go home?* Feeding well, gaining weight for 3 days, temperature stable for 3 days, no danger signs, mother confident.

On the ward round

1. **"Is this baby preterm, SGA, or both?"** Plot weight against gestation (INTERGROWTH-21st or Fenton); <10th centile = SGA.
2. **"What is his temperature and how are we keeping him warm?"** Give the number and the method (KMC hours or incubator).
3. **"What are his total fluids in mL/kg/day, and how are they split?"** For example "120 mL/kg/day: 60 enteral, 60 IV".
4. **"When will you stop caffeine?"** About 34–35 weeks PMA and 5–7 days apnoea-free.
5. **"Why is he on antibiotics, and when will you stop?"** Risk factor or sign; stop at 48 h if well and culture negative; 7–10 days if sepsis.
6. **"Does he need ROP screening?"** Yes if <32 weeks or <1.5 kg (wider in LMICs) or unmonitored oxygen; first examination at 4 weeks or 31 weeks PMA.
7. **"What will you tell the mother before discharge?"** KMC, exclusive breastfeeding, danger signs, weekly weighing, immunisation, follow-up for eyes, hearing, development.

Mnemonics

Problems of prematurity: "TIRED BABY"

- **T**emperature (hypothermia)
- **I**nfection
- **R**espiratory (RDS, apnoea, BPD)
- **E**yes (ROP)
- **D**igestive (feeding, NEC)
- **B**rain (IVH, PVL)
- **A**naemia
- **B**ilirubin (jaundice)
- **Y**our sugar (hypoglycaemia)

Fluids: "60-90-120-150" (day 1-2-3-4, mL/kg/day).

KMC: "Skin, Sling, Suckle, Stay": skin-to-skin, secure binder, breastfeed in position, stay day and night.

OSCE tips

- **Counselling a mother on KMC** is a common OSCE station: introduce, explain why (warmth, breastfeeding, infection, bonding), demonstrate position and binder, how long, sleeping position, danger signs, hygiene, who can help, check understanding.
- **Calculating feeds** is a favourite: show your working (weight × mL/kg/day ÷ number of feeds).
- **Examination station**: keep the baby warm, plot weight on a preterm chart, state gestation and weight-for-gestation, and do a Ballard score if dates are uncertain.

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SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define preterm birth and its subgroups.	Born **before 37 completed weeks**. Extremely preterm **below 28 weeks**; very preterm **28 to below 32**; moderate-late preterm **32 to below 37** (late preterm 34-36+6).
2	Define LBW, VLBW and ELBW.	Birth weight **below 2,500 g** (LBW), **below 1,500 g** (VLBW), **below 1,000 g** (ELBW), regardless of gestation.
3	What is SGA, and what is its main early risk?	Birth weight **below the 10th centile** for gestation and sex. Main early risks: **hypoglycaemia** (low glycogen stores), **polycythaemia**, hypothermia, asphyxia.
4	WHO 2022: who should receive Kangaroo Mother Care, when, and for how long?	**All preterm or LBW infants**, started **as soon as possible after birth** (immediate KMC), for **8-24 hours per day** (as many hours as possible).
5	WHO temperature classification in the newborn.	Normal **36.5-37.5 C**; cold stress **36.0-36.4**; moderate hypothermia **32.0-35.9**; severe hypothermia **below 32 C**.
6	Total daily fluid volumes for a newborn on days 1-4 (WHO Pocket Book).	Day 1 **60**, day 2 **90**, day 3 **120**, day 4+ **150 mL/kg/day** (up to 180 when tolerating feeds). Count IV + enteral together.
7	How do you start feeds in a VLBW (below 1.5 kg) baby on day 1?	**10 mL/kg/day** of expressed breast milk (or 2-4 mL every 2 h by NGT if well) with the rest (~50 mL/kg/day) as IV 10% dextrose; advance by 1-2 mL per feed per day (Pocket Book) or **up to 30 mL/kg/day** (WHO 2022).
8	Order of preference of milk for a preterm baby (WHO 2022).	**Mother's own milk** → **donor human milk** (if a safe milk bank exists) → **preterm formula** only if neither is available.
9	At what gestation does suck-swallow-breathe coordination usually mature?	About **32-34 weeks**. Before this, feed by **nasogastric/orogastric tube**; then cup/paladai; then breast.
10	Define apnoea of prematurity.	Cessation of breathing for **20 seconds or more**, or shorter with **bradycardia (HR below 100)** or desaturation, in a preterm baby (usually below 34 weeks), with no other cause found.
11	Caffeine citrate regimen for apnoea of prematurity (WHO Pocket Book).	**Loading 20 mg/kg** oral or IV over 30 min, then **5 mg/kg once daily** from 24 h later (may increase in 5 mg/kg steps to max 20 mg/kg/day). Continue to about **34-35 weeks PMA** and 5-7 days apnoea-free.
12	Aminophylline dose if caffeine is unavailable.	**6 mg/kg IV** over 20 min loading, then **2.5 mg/kg 12-hourly** in week 1 (4 mg/kg 12-hourly in weeks 2-4). Narrower safety margin than caffeine.
13	Clinical features of RDS.	Preterm baby with **tachypnoea, expiratory grunt, intercostal/subcostal recession, nasal flaring and cyanosis**, starting **within 4 h of birth**, worsening over 24-48 h and improving after 48-72 h.
14	Mechanism of RDS.	**Surfactant deficiency** (immature type II pneumocytes) → alveolar collapse at end-expiration → atelectasis, V/Q mismatch, hypoxaemia, increased work of breathing.
15	CPAP settings for a preterm with RDS.	Bubble CPAP starting at **5-6 cm H2O** (up to 7-8), FiO2 titrated to **SpO2 90-95%**, correct prong size, NGT on free drainage, nasal care.
16	SpO2 target in preterm babies on oxygen, and why.	**90-95%**. Below this: hypoxic injury and death; above: **retinopathy of prematurity** and lung injury (BPD).

#	Question	Answer
17	Who should be screened for ROP and when?	International: **below 31-32 weeks or below 1.5 kg**; LMIC programmes often wider (up to 34 weeks / 2.0 kg, or any unmonitored oxygen). First exam at **4 weeks after birth or 31 weeks PMA, whichever is later**.
18	Signs of necrotising enterocolitis.	**Abdominal distension**, feed intolerance, **bilious aspirates/vomit**, **blood in stool**, abdominal tenderness/discoloration, plus apnoea, lethargy, temperature instability. AXR: **pneumatosis intestinalis**.
19	Management of suspected NEC.	**Stop feeds**, NGT on free drainage, IV fluids, **ampicillin + gentamicin + metronidazole** (10 days), oxygen if needed, transfuse if Hb below 10 g/dL, AXR; surgical review if perforation.
20	Iron and vitamin D supplementation in preterm/LBW babies.	**Iron 2-4 mg/kg/day** elemental from **2 weeks to 6 months**; **vitamin D 400 IU/day** once on full feeds (WHO).
21	WHO admission threshold for special newborn care.	All babies **below 2.0 kg or below 35 weeks** should be admitted to a special care unit. UCG 2023: refer babies **below 1.5 kg or below 32 weeks** urgently from HC III/IV.
22	Discharge criteria for a LBW baby.	No danger signs; **feeding well** (breast/cup) and **gaining weight on 3 consecutive days**; temperature normal for **3 consecutive days**; mother confident; vaccines and follow-up arranged.
23	How often should a LBW baby be reviewed after discharge?	**Weekly** for weight and feeding until **3 kg** (WHO); growth and development plotted on **corrected age** until 2 years.
24	Do preterm babies receive vaccines at corrected or chronological age?	**Chronological age.** BCG and OPV0 at birth/before discharge; UNEPI 6-10-14 week vaccines from 6 weeks after birth.
25	Why do preterm babies lose heat so easily?	Large **surface-area-to-weight ratio**, thin permeable skin (**evaporative** loss), little **brown fat** and subcutaneous fat, extended posture, cannot shiver.
26	Calculate feeds: 1.5 kg baby on 150 mL/kg/day, 3-hourly.	1.5 × 150 = **225 mL/day** ÷ 8 feeds = about **28 mL 3-hourly**.
27	Approximate KMC position.	Baby in **nappy, hat and socks**, upright **skin-to-skin between the mother's breasts**, head turned to one side and slightly extended, secured with a binder, covered by mother's clothes; mother sleeps propped up.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A baby is born at a HC IV at 33 weeks by early scan, weight 1.70 kg. At 1 hour of age she is pink, RR 52/min, no grunting, temperature 35.8 C (axillary). The mother is well and wants to hold her. What is the MOST appropriate next step?

A Place her under a 100 W bulb and recheck the temperature after 4 hours
B Give her a warm bath to raise the temperature quickly
C Start skin-to-skin Kangaroo Mother Care now, cover the head, and recheck the temperature in 1 hour
D Refer immediately to the regional referral hospital without warming
E Give 10% dextrose 2 mL/kg IV before any warming

2. A 1.2 kg baby born at 30 weeks is on day 2. He is stable in room air, abdomen soft, passed meconium. Which feeding plan is MOST appropriate today?

A Total fluids 90 mL/kg/day: small 2-hourly NGT feeds of expressed breast milk with the remainder as IV fluid
B Nil by mouth until day 7, IV 10% dextrose 150 mL/kg/day
C Direct breastfeeding on demand only
D Formula 20 mL 3-hourly by bottle
E IV fluids 90 mL/kg/day plus full NGT feeds of 150 mL/kg/day

3. A 29-week baby (1.1 kg) on day 2 has three episodes in 6 hours where breathing stops for about 25 seconds with HR dropping to 80/min; she recovers with gentle stimulation. Temperature 36.8 C, glucose 3.4 mmol/L, no signs of sepsis, on ampicillin and gentamicin for maternal PPROM. What is the MOST appropriate drug treatment?

A Phenobarbital 20 mg/kg IV
B Diazepam 0.3 mg/kg IV
C Adrenaline 0.01 mg/kg IV
D Caffeine citrate 20 mg/kg loading dose, then 5 mg/kg daily
E Aminophylline 20 mg/kg IV, then 20 mg/kg daily

4. A baby born at 31 weeks by emergency CS for severe pre-eclampsia (mother received one dose of dexamethasone) develops grunting, subcostal recession and RR 74/min at 2 hours of age. SpO2 is 86% in air. Bubble CPAP is available. What is the MOST appropriate management?

A Oxygen by head box at 10 L/min and observe
B Start bubble CPAP 5-6 cm H2O with blended oxygen to SpO2 90-95%, IV fluids, and ampicillin + gentamicin after a blood culture
C Give furosemide 1 mg/kg IV for lung fluid
D Start NGT feeds of 150 mL/kg/day to prevent hypoglycaemia
E Wait for a chest X-ray before any respiratory support

5. A 32-week, 1.4 kg baby on day 8 who has been stable and feeding well on NGT suddenly develops abdominal distension, bilious aspirates and fresh blood in the stool. She has had two new apnoeas. What is the MOST appropriate immediate step?

A Increase the caffeine dose and continue feeds
B Change to formula feeds
C Give a glycerine suppository and continue feeds
D Reduce the feeds by half and recheck in 24 hours
E Stop feeds, put the NGT on free drainage, start IV fluids and ampicillin + gentamicin + metronidazole, and request an abdominal X-ray

6. A term baby weighing 2.1 kg (below the 3rd centile) is born to a mother with chronic hypertension. At 3 hours he is alert, sucking eagerly, temperature 36.7 C. Which complication should you MOST actively screen for in the first 24 hours?

A Respiratory distress syndrome
B Hypoglycaemia
C Retinopathy of prematurity
D Necrotising enterocolitis
E Apnoea of prematurity

7. You are calculating fluids for a 1.6 kg baby on day 3 of life. He tolerates 8 mL of expressed breast milk every 2 hours by NGT. What should the IV fluid rate be?

A 8 mL/h
B 1 mL/h
C 6 mL/h
D 4 mL/h (total 192 mL/day minus feeds 96 mL/day = 96 mL/day IV)
E No IV fluid is needed

8. A 28-week baby has been on bubble CPAP with 100% oxygen (no blender) for 10 days at a district hospital. SpO2 has been 98-100% throughout. Which long-term complication is this practice MOST likely to cause?

A Retinopathy of prematurity
B Sensorineural hearing loss
C Cerebral palsy from kernicterus
D Hypothyroidism
E Developmental dysplasia of the hip

9. A mother of a 1.5 kg baby born at 33 weeks asks how long each day she should keep the baby in the Kangaroo position. According to WHO 2022, what is the BEST answer?

A One hour, twice a day
B Only at night
C Only after the baby reaches 2 kg
D Only while breastfeeding
E As many hours as possible, ideally continuously day and night (at least 8 and up to 24 h/day), starting now

10. A 34-week, 2.0 kg baby on day 6 is feeding well at the breast, gaining weight for 2 days, temperature 36.6-37.0 C for 3 days in KMC, no danger signs. The mother is keen to go home today; she lives 40 km away. What is the MOST appropriate decision?

A Discharge today with no follow-up because he is over 2 kg

B Keep in hospital until he reaches 2.5 kg

C Keep one more day to confirm weight gain on 3 consecutive days, then discharge with KMC at home and weekly weighing until 3 kg

D Discharge today on formula top-ups to speed weight gain

E Transfer to an incubator for 3 more days

11. Which statement about immunisation of a 30-week preterm baby is CORRECT according to WHO/UNEPI practice?

A BCG and OPV0 should be given at birth or before discharge, and later vaccines by chronological age

B All vaccines should be delayed until the corrected age of 40 weeks

C BCG is contraindicated below 2.5 kg

D The 6-week vaccines should be given at 6 weeks corrected age

E Preterm babies should not receive rotavirus vaccine

12. A 32-week baby on day 4 in the newborn unit has temperature 35.2 C despite a warm room, is feeding poorly and has a new apnoea. Glucose is 2.1 mmol/L. After giving 10% dextrose 2 mL/kg IV, what is the MOST important next step?

A Increase the caffeine dose

B Take a blood culture and start ampicillin + gentamicin

C Give paracetamol

D Stop all feeds for 72 hours

E Reassure the mother that this is normal for prematurity

13. A 1.3 kg, 31-week baby is born at a HC III. The midwife has dried and warmed her; she is breathing with mild recession, SpO2 not available. What is the MOST appropriate plan according to UCG 2023?

A Keep at HC III and discharge home after 24 hours if feeding

B Bathe and dress the baby before referral

C Refer urgently to hospital in skin-to-skin KMC with the mother, after feeding or giving expressed breast milk and a first antibiotic dose if indicated

D Wait until she is 1.5 kg before referral

E Give formula by bottle and refer the next morning

14. Which pathophysiological feature BEST explains why respiratory distress syndrome improves after 48-72 hours?

A Absorption of lung fluid by lymphatics

B Closure of the ductus arteriosus

C Clearance of meconium from the airways

D Endogenous surfactant production increases after birth

E Maturation of brainstem respiratory control

15. A 6-week-old baby born at 30 weeks (birth weight 1.25 kg) attends the follow-up clinic. Weight is 2.1 kg, feeding exclusively on breast milk. She is on vitamin D. What is MOST important to add today?

- **A** Cow's milk to increase weight gain
- **B** Ceftriaxone prophylaxis
- **C** Delay her 6-week vaccines until 6 weeks corrected age
- **D** Stop KMC because she is above 2 kg
- **E** Oral iron 2-4 mg/kg/day until 6 months, give the 6-week vaccines, and arrange ROP and hearing follow-up

Answers & explanations

1. C · Start skin-to-skin Kangaroo Mother Care now, cover the head, and recheck the temperature in 1 hour

C is right: she has moderate hypothermia (32.0-35.9 C) but is otherwise stable; WHO 2022 recommends immediate KMC for preterm/LBW babies, and skin-to-skin is the most effective way to rewarm and to start breastfeeding. Recheck within an hour. A is wrong: bulbs are unreliable, can burn, and 4 hours is too long. B is wrong: bathing a small cold baby worsens heat loss (evaporation). D is wrong: a 1.7 kg baby can be cared for at HC IV (UCG refers below 1.5 kg or below 32 weeks), and referring a cold baby without warming is dangerous. E is wrong: there is no hypoglycaemia documented; check glucose and feed, but warming is the priority and does not need to wait for IV dextrose.

2. A · Total fluids 90 mL/kg/day: small 2-hourly NGT feeds of expressed breast milk with the remainder as IV fluid

A is right: day 2 total fluids are 90 mL/kg/day (WHO Pocket Book). A VLBW baby who is stable should receive expressed breast milk by NGT from day 1, advancing gradually, with the balance IV. B is wrong: prolonged nil by mouth increases infection risk and delays gut maturation, and 150 mL/kg/day IV on day 2 risks fluid overload (IV should not exceed about 100 mL/kg/day). C is wrong: a 30-week baby cannot coordinate suck-swallow-breathe (matures about 32-34 weeks). D is wrong: formula is the last choice and bottles are discouraged; 20 mL 3-hourly is 133 mL/kg/day, too much for day 2. E is wrong: it double-counts fluids (240 mL/kg/day total), a classic error.

3. D · Caffeine citrate 20 mg/kg loading dose, then 5 mg/kg daily

D is right: these are typical apnoeas of prematurity in a 29-week baby; temperature and glucose are normal and sepsis is being covered. WHO 2022 strongly recommends caffeine for treatment of apnoea; the Pocket Book dose is caffeine citrate 20 mg/kg load, then 5 mg/kg once daily. A is wrong: phenobarbital is for seizures and would worsen apnoea. B is wrong: diazepam depresses respiration. C is wrong: adrenaline is for HR below 60 despite ventilation and compressions during resuscitation. E is wrong: aminophylline is the second choice and these are caffeine-like doses (aminophylline load is 6 mg/kg, maintenance 2.5 mg/kg 12-hourly); 20 mg/kg aminophylline would be toxic.

4. B · Start bubble CPAP 5-6 cm H2O with blended oxygen to SpO2 90-95%, IV fluids, and ampicillin + gentamicin after a blood culture

B is right: this is likely RDS (preterm, onset within 4 h, grunting and recession). WHO 2022 strongly recommends CPAP for RDS, with oxygen titrated to SpO2 90-95%; IV fluids and antibiotics are given because pneumonia/sepsis cannot be excluded clinically. A is wrong: high-flow unblended oxygen without pressure does not recruit alveoli and risks hyperoxia (ROP). C is wrong: diuretics have no role in RDS. D is wrong: a baby in respiratory distress should initially be nil by mouth or on small trophic feeds, and 150 mL/kg/day on day 1 is far too much. E is wrong: a CXR helps but must not delay treatment of a hypoxic, grunting baby.

5. E · Stop feeds, put the NGT on free drainage, start IV fluids and ampicillin + gentamicin + metronidazole, and request an abdominal X-ray

E is right: distension, bilious aspirates, bloody stool and new apnoeas in a preterm on feeds suggest necrotising enterocolitis. Management is nil by mouth, gastric decompression, IV fluids, triple antibiotics (WHO Pocket Book: 10 days), AXR (pneumatosis, free air), and surgical review if perforation. A is wrong: the apnoeas are a sign of systemic illness (NEC/sepsis), not simple apnoea of prematurity. B is wrong: formula increases NEC risk. C is wrong: continuing feeds in NEC is dangerous. D is wrong: half feeds still stresses an ischaemic gut; feeds must stop.

6. B · Hypoglycaemia

B is right: a term SGA baby has small glycogen and fat stores and is at high risk of hypoglycaemia (and polycythaemia); feed early and check glucose before feeds in the first 24 h. A is wrong: RDS is a problem of preterm surfactant deficiency, not of a term growth-restricted baby. C is wrong: ROP affects very preterm babies exposed to oxygen. D is wrong: NEC is mainly a preterm problem. E is wrong: apnoea of prematurity is, by definition, a problem of preterm babies.

7. D · 4 mL/h (total 192 mL/day minus feeds 96 mL/day = 96 mL/day IV)

D is right: day 3 total fluids = 120 mL/kg/day × 1.6 = 192 mL/day. Feeds = 8 mL × 12 = 96 mL/day. IV = 192 − 96 = 96 mL/day = 4 mL/h. Showing the working is the key skill. A is wrong: 8 mL/h is 192 mL/day of IV on top of the feeds (double counting). B is wrong: 1 mL/h (24 mL/day) under-hydrates him. C is wrong: 6 mL/h (144 mL/day IV) plus 96 mL of feeds gives 240 mL/day, i.e. the day-4 volume of 150 mL/kg/day. E is wrong: feeds alone give only 60 mL/kg/day on day 3.

8. A · Retinopathy of prematurity

A is right: prolonged hyperoxia (SpO2 98-100% on unblended oxygen) in a very preterm baby drives abnormal retinal vessel growth: ROP. Target SpO2 90-95% and screen this baby. B is wrong: hearing loss is linked to hyperbilirubinaemia, aminoglycosides, meningitis and hypoxia, not to high SpO2. C is wrong: kernicterus results from bilirubin, not oxygen. D is wrong: hypothyroidism is not caused by oxygen. E is wrong: DDH is linked to breech, female sex and family history.

9. E · As many hours as possible, ideally continuously day and night (at least 8 and up to 24 h/day), starting now

E is right: WHO 2022 recommends KMC for all preterm/LBW infants, started as soon as possible after birth, for 8-24 hours per day (as many hours as possible). A and D are wrong: short sessions give much less benefit. B is wrong: KMC is continuous, day and night, with relatives able to take turns. C is wrong: waiting until 2 kg misses the period of greatest benefit; immediate KMC reduces mortality.

10. C · Keep one more day to confirm weight gain on 3 consecutive days, then discharge with KMC at home and weekly weighing until 3 kg

C is right: WHO/UCG discharge criteria include weight gain on 3 consecutive days, normal temperature for 3 days, feeding well, no danger signs and a confident mother; he has weight gain for only 2 days. Follow-up should be weekly until 3 kg. A is wrong: a weight threshold alone is not a discharge criterion, and follow-up is essential. B is wrong: there is no need to wait for 2.5 kg once criteria are met. D is wrong: formula top-ups undermine exclusive breastfeeding and add infection risk. E is wrong: a stable baby in KMC does not need an incubator.

11. A · BCG and OPV0 should be given at birth or before discharge, and later vaccines by chronological age

A is right: WHO advises that LBW babies receive scheduled birth vaccines at birth or by discharge and later doses by chronological age. B and D are wrong: corrected age is used for growth and development, not immunisation; delaying leaves the baby unprotected. C is wrong: low weight is not a contraindication to BCG in the UNEPI programme. E is wrong: rotavirus vaccine is given to preterm babies by chronological age when clinically stable.

12. B · Take a blood culture and start ampicillin + gentamicin

B is right: hypothermia, poor feeding, apnoea and hypoglycaemia together in a preterm baby are classic non-specific signs of sepsis. After correcting glucose, warm the baby, take a culture and start empirical ampicillin + gentamicin. A is wrong: a new apnoea with other danger signs must not be blamed on prematurity. C is wrong: there is no fever, and antipyretics are not used in neonates. D is wrong: feeds may need to be reduced while unwell, but a blanket 72-hour stop is not indicated and risks further hypoglycaemia. E is wrong: these are danger signs.

13. C · Refer urgently to hospital in skin-to-skin KMC with the mother, after feeding or giving expressed breast milk and a first antibiotic dose if indicated

C is right: UCG 2023 classifies a baby below 1.5 kg or below 32 weeks as 'very small baby: refer urgently to hospital for special care' and stresses extra warmth during referral; KMC during transport is the safest method, and prevention of hypoglycaemia and early antibiotics improve survival. A is wrong: very small babies need specialised care and should not be discharged early. B is wrong: bathing a small baby causes hypothermia. D is wrong: delay increases risk of death. E is wrong: bottles and formula are not preferred, and delaying referral is dangerous.

14. D · Endogenous surfactant production increases after birth

D is right: RDS is due to surfactant deficiency; birth triggers increased surfactant production, so the disease is self-limiting if the baby is supported for the first 2-3 days. A is wrong: lung fluid clearance explains transient tachypnoea of the newborn. B is wrong: ductal closure is not the mechanism of recovery in RDS (a PDA may even worsen it). C is wrong: that is meconium aspiration syndrome. E is wrong: brainstem maturation explains the resolution of apnoea of prematurity.

15. E · Oral iron 2-4 mg/kg/day until 6 months, give the 6-week vaccines, and arrange ROP and hearing follow-up

E is right: human-milk-fed preterm babies need enteral iron (WHO: 2-4 mg/kg/day from 2 weeks to 6 months), vaccines by chronological age, and screening for ROP (born below 32 weeks) and hearing. A is wrong: cow's milk is unsuitable before 12 months and displaces breast milk. B is wrong: there is no indication for antibiotic prophylaxis. C is wrong: immunisation uses chronological age. D is wrong: KMC can continue at home until the baby no longer tolerates it (around 2.5 kg or term).

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: A 1.4 kg baby with breathing difficulty at Mengo

You are the junior clerk on the Mengo Hospital newborn unit. A baby boy is admitted from labour ward at 1 hour of age. He was born vaginally at 31 weeks (early scan) after preterm labour; membranes ruptured 26 hours before birth; the mother had a temperature of 38.4 C in labour and received one dose of dexamethasone. Birth weight 1.40 kg. He is grunting with subcostal recession, RR 76/min, axillary temperature 35.6 C, SpO2 85% in air, HR 168/min.

1. What are your immediate priorities in the first 5 minutes?

> **Warmth**: radiant warmer/incubator or skin-to-skin if stable enough; hat. **Breathing**: position, oxygen to SpO2 90-95%, prepare **bubble CPAP 5-6 cm H2O** with blended oxygen. **IV access** and **glucose check**. Call the senior. Draw a **blood culture** and start antibiotics. Minimal handling.

2. What history and examination findings will you document, and why?

> Gestation (31 weeks, early scan: reliable), **steroids** (only 1 dose: incomplete), **ROM 26 h and maternal fever** (sepsis risk), resuscitation at birth, maternal HIV/VL, syphilis, HBsAg. Exam: RR, grunting, recession, air entry, SpO2, CRT, temperature, tone, fontanelle, abdomen, Ballard to confirm gestation, weight on preterm chart (1.40 kg at 31 weeks = AGA).

3. What is your differential diagnosis for his respiratory distress?

> **RDS** (preterm, onset within 4 h, incomplete steroids) is most likely, but **congenital pneumonia/early-onset sepsis** is very possible (ROM 26 h, maternal fever) and cannot be excluded clinically. Others: transient tachypnoea (less likely in a 31-weeker after vaginal birth), pneumothorax, hypothermia, congenital heart disease.

4. What investigations will you request and what will change management?

> **Glucose** (now and 3-6-hourly), **blood culture**, **FBC and CRP** (at 18-24 h), **CXR** (ground-glass and air bronchograms in RDS; patchy in pneumonia; exclude pneumothorax). LP later if culture positive or he has seizures. A positive culture or rising CRP means a full course (7-10 days; 21 days if meningitis).

5. Write the management plan with doses (he weighs 1.40 kg).

> 1. Bubble **CPAP 5-6 cm H2O**, FiO2 to SpO2 **90-95%**. 2. **Ampicillin 50 mg/kg = 70 mg IV 12-hourly** + **gentamicin 3 mg/kg = 4.2 mg IV once daily** (WHO LBW dose; UCG 4 mg/kg = 5.6 mg). 3. Fluids day 1 **60 mL/kg/day = 84 mL/day = 3.5 mL/h** of 10% dextrose; start **trophic EBM 10 mL/kg/day** once stable. 4. **Caffeine citrate 20 mg/kg = 28 mg** load then **7 mg daily** (consider prevention below 34 weeks / treat apnoea). 5. Vitamin K 0.5 mg IM, chlorhexidine cord care, eye ointment. 6. Monitor T, RR, SpO2, HR, glucose; strict fluid chart; weigh daily.

6. On day 3 he is in room air, culture negative, CRP normal. On day 9 he develops apnoeas, temperature 35.4 C and a distended abdomen with bilious aspirates. What do you do?

> Think **late-onset sepsis and NEC**. Stop feeds, NGT on free drainage, IV fluids, **blood culture**, start **ampicillin (or cloxacillin if skin/line sepsis) + gentamicin 5 mg/kg (now 7.5 mg/kg after week 1 per WHO) + metronidazole**; or second line per unit (ceftriaxone/meropenem) if hospital-acquired. AXR for pneumatosis/free air; Hb and transfuse if below 10 g/dL; surgical review if perforation. Warm and check glucose.

7. What follow-up does he need after discharge?

> Weekly weighing until **3 kg**; KMC at home; exclusive breastfeeding; **iron** 2-4 mg/kg/day from 2 weeks to 6 months and vitamin D; **ROP screen** at 4 weeks of age (born below 32 weeks, exposed to oxygen); hearing screen (gentamicin, sepsis); development review using corrected age; vaccines by chronological age.

> **Take-away:** In a preterm baby with respiratory distress and sepsis risk factors, treat RDS (CPAP, safe oxygen) and sepsis together, count total fluids carefully, and remember that a new apnoea later in the stay is sepsis or NEC until proven otherwise.

Case 2: Twins at a HC IV: who stays and who goes?

You are on rotation at a HC IV in Mpigi. A 24-year-old para 2 has just delivered twins at about 34 weeks (LNMP uncertain, no scan). Twin 1 (girl) weighs 1.95 kg; twin 2 (boy) weighs 1.35 kg. Both cried at birth. At 30 minutes, twin 2's temperature is 35.4 C and he is not attaching to the breast; twin 1 is 36.4 C and suckling.

1. How do you classify each twin and what does UCG 2023 advise?

> Twin 1: **LBW (1.5-2.5 kg)** 'small baby': care at HC IV with KMC, breastfeeding support, cup feeds if needed, daily assessment. Twin 2: **VLBW (below 1.5 kg)** 'very small baby': **refer urgently to hospital** with extra warmth during referral. Twins should be discharged together only when both are fit.

2. What will you do for twin 2 before he leaves?

> **KMC** on the mother or a relative (or both twins on the mother), hat and socks; **glucose** check if possible and **expressed colostrum** by cup/spoon or NGT (or 10% dextrose 2 mL/kg IV if hypoglycaemic and IV possible); **vitamin K**, eye ointment, chlorhexidine cord care; consider the first dose of **ampicillin + gentamicin** if risk factors; write a **referral note** (times, weights, temperatures, treatments).

3. How will you estimate gestation, given uncertain dates?

> **New Ballard score** on both (neuromuscular + physical maturity). The difference between twins also helps: twin 2 may be **SGA** (discordant growth, e.g. placental insufficiency) rather than more preterm. Plot weights on INTERGROWTH-21st charts.

4. Twin 1 stays. What feeding and thermal plan will you write?

> **Continuous KMC** (8-24 h/day); breastfeed on demand at least 8-12 times/day; if not suckling well, **expressed breast milk by cup**: day 1 60 mL/kg/day = **about 120 mL/day = 15 mL 3-hourly**, increasing daily to 150 mL/kg/day. Check temperature 4-6-hourly, glucose before feeds in the first 24 h, daily weight, jaundice check.

5. When can twin 1 be discharged and what counselling will you give?

> When **feeding well**, **gaining weight 3 consecutive days**, **temperature normal 3 days**, mother confident, and (UCG) when **both twins are fit to go home**. Counsel: KMC at home, exclusive breastfeeding, hygiene and cord care, **danger signs** (not feeding, cold, fast breathing, convulsions, yellow palms/soles), BCG/OPV0 before discharge, weekly weighing until 3 kg.

6. What antenatal opportunities were missed, and what will you tell the mother for the future?

> No early scan (accurate dating), no plan for twin delivery at a hospital with a newborn unit, possibly no **antenatal steroids**. Advise early ANC, early scan, birth spacing (family planning), delivery planning in hospital for future multiple or high-risk pregnancy.

> **Take-away:** Classify every small baby by weight and gestation: babies 1.5-2.5 kg can be cared for at HC IV with KMC and cup feeding, while babies below 1.5 kg or below 32 weeks need urgent, warm referral after

glucose and infection prevention.

Case 3: Weight loss and poor feeding in a late preterm baby

A 36-week baby girl born by elective CS (birth weight 2.40 kg) is brought back to Mengo outpatients on day 5. The mother says the baby 'sleeps all the time' and feeds for only a few minutes. She weighs 2.10 kg today. Temperature 36.3 C, RR 48, HR 150, mildly jaundiced to the abdomen, palms and soles not yellow. Mother HIV-negative, blood group O positive.

1. What is her percentage weight loss and is it acceptable?

> (2.40 – 2.10) ÷ 2.40 = **12.5%**. More than **10%** is excessive and suggests **inadequate milk intake** (late preterm babies are sleepy with weak sucks).

2. What focused history and examination will you do?

> Feeds per 24 h (target 8-12), duration, audible swallowing, wet nappies (6 or more/day by day 5) and stool colour (transitioning to yellow by day 4-5), any other fluids. Breast exam (engorgement, cracked nipples). Baby: hydration (fontanelle, mucosa), jaundice extent, temperature, signs of sepsis (lethargy, poor tone), **observe a breastfeed** (attachment, suckling).

3. Which bedside and lab tests are useful?

> **Glucose**, **temperature**, **serum (or transcutaneous) bilirubin** (late preterm + poor feeding = higher risk; compare to 35-37 week thresholds), **serum sodium** if available (hypernatraemic dehydration), and FBC/CRP/blood culture if any sign of sepsis.

4. How will you manage her?

> Admit for feeding support: correct positioning and attachment, feed **at least every 3 h** (wake her), **express breast milk** after feeds and give by cup to reach about **150 mL/kg/day** (≈ 360 mL/day ≈ 45 mL 3-hourly by birth weight; start lower and increase), skin-to-skin/KMC for warmth (36.3 C = cold stress), phototherapy if bilirubin above threshold, treat hypoglycaemia if present, sepsis work-up if unwell.

5. What is the key lesson about late preterm babies?

> **Late preterm (34-36+6) babies look like term babies but behave like preterm ones**: weak suck, sleepiness, hypothermia, hypoglycaemia and jaundice. Do not discharge early; check feeding, weight and jaundice at a **2-3-day review**.

Take-away: Late preterm babies are at risk of poor feeding, excess weight loss, hypothermia and jaundice; calculate percentage weight loss, observe a feed, and support expressed breast milk feeding before problems become emergencies.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.